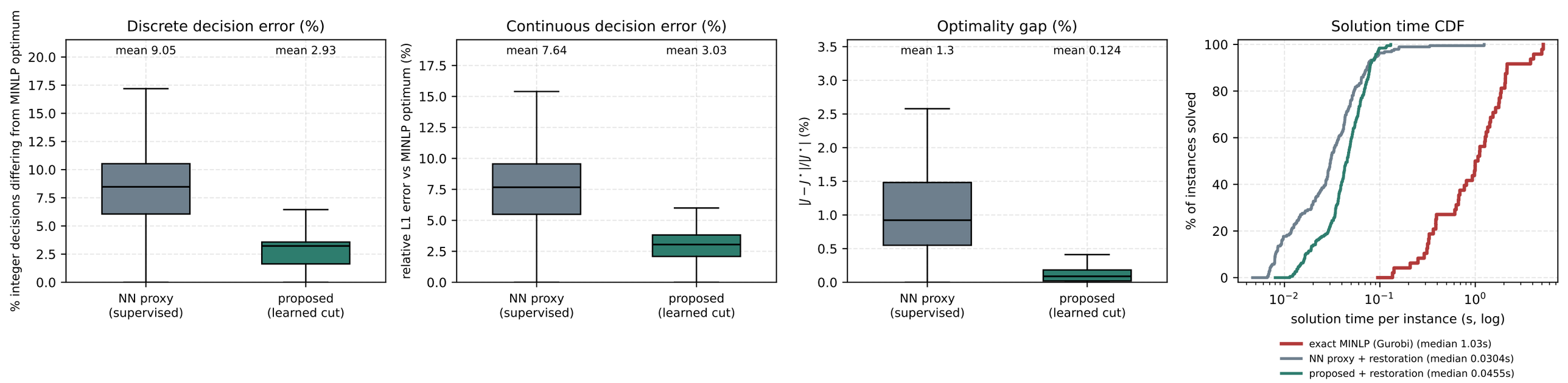
Discrete decision error (%)
% integer decisions differing from MINLP optimum
mean 9.05
mean 2.93
NN proxy (supervised)
proposed (learned cut)
Continuous decision error (%)
relative L1 error vs MINLP optimum (%)
mean 7.64
mean 3.03
NN proxy (supervised)
proposed (learned cut)
Optimality gap (%)
$|J - J^\star|/|J^\star|$ (%)
mean 1.3
mean 0.124
NN proxy (supervised)
proposed (learned cut)
Solution time CDF
% of instances solved
solution time per instance (s, log)
exact MINLP (Gurobi) (median 1.03s)
NN proxy + restoration (median 0.0304s)
proposed + restoration (median 0.0455s)